

ORCA

AI Cost & Scalability Analysis

Measured AI Inference Costs and Scenario-Based Scale Projection

1. Scope of Analysis

This document presents the measured and estimated AI inference costs associated with ORCA, specifically the Gemini model used for processing user queries and generating responses.

Scope: The figures presented here represent AI model costs only and should not be interpreted as the complete operational cost of the ORCA platform.

2. AI Cost Mechanism

Each user interaction with ORCA involves two primary AI operations:

Input Processing

- Processing the user's query, system instructions, and available tool/context information. These contribute to input token consumption.

Response Generation

- Generating the AI response, which contributes to output token consumption.

The Dify monitoring dashboard records token consumption and the corresponding Gemini API cost. The figures below are therefore measured platform data rather than theoretical calculations.

3. Measured AI Usage & Cost

Monitoring Period	Messages	Tokens Used	Measured AI Cost
27 September 2026	73	669,000	\$0.2355 (~₹22.61)
Previous 7 Days	158	1,341,000	\$0.4666 (~₹44.79)
Previous 30 Days	163	1,374,000	\$0.4757 (~₹45.67)

The usage data indicates that the majority of ORCA's recorded activity occurred during the most recent seven-day period, reflecting the project's recent development and testing phase.

4. Average AI Cost per User Interaction

Configuration	Approx. Cost / Message
Before Memory Integration	₹0.26
After Memory Integration	₹0.31

The introduction of conversational memory increased the observed average cost per interaction from approximately ₹0.26 to ₹0.31, representing an increase of roughly 19%. This increase is associated with the additional conversational context that the model processes to maintain continuity across interactions.

5. Projected Cost at User Scale

Note: The following figures are scenario-based estimates, not measured expenditure.

The projection uses the following assumptions:

- Approximately 4 million marine fishermen in India, based on the CMFRI Marine Fisheries Census, 2010.
- Approximately 3 ORCA interactions per active user per day, assuming short operational queries such as safety conditions or fishing-zone information.
- An assumed first-year adoption of approximately 125,000 active users.

Metric	Estimated Cost
AI cost per fisherman / day	~₹0.93
AI cost per fisherman / month	~₹27.90
AI cost for 125,000 users / day	~₹1,16,250
AI cost for 125,000 users / month	~₹34.88 lakh

These figures should be treated as planning estimates and will vary with actual user adoption, interaction frequency, token consumption, model configuration, and conversational context.

6. Cost Components Not Included

Bhashini: Used for speech-to-text, text-to-speech and language translation. A usage-based cost figure was not available in the current monitoring data.

Sea-Route Planning Service: The route-planning service is hosted separately on Render; its infrastructure and compute costs are not included in this analysis.

Node.js Backend: The server responsible for connecting ORCA's services is not included in the current cost measurement.

External Data Sources: Weather, ocean, fishing-zone and chlorophyll datasets currently used by ORCA are sourced from publicly available data sources. Their availability as public data should not be confused with the infrastructure cost of retrieving, processing and serving that data.

7. Key Financial Takeaway

The current analysis establishes a baseline AI inference cost of approximately ₹0.26–₹0.31 per user interaction, based on ORCA's measured usage.

At the assumed scale of 125,000 active users, AI inference alone could represent approximately ₹34.9 lakh per month, under the stated usage assumptions.

Important: This is not the total cost of operating ORCA. A complete production-cost model would additionally need to account for backend infrastructure, route-planning compute, Bhashini usage, databases, monitoring, storage, networking and other operational services.

